Q8: The chemical formula of the precipitate formed when $CaCl_2$ solution is allowed to react with Na_2CO_3 solution is:

A) $CaCO_3$ B) $CaCl_2$ C) NaCl D) Na_2CO_3

Q9: The data given in the table beside are collected to determine the empirical formula of magnesium oxide. Find the mass percent of Mg in MgO.

Mass of crucible	6.00 g
Mass of crucible + Mg	6.218 g
Mass of crucible + Mg and O	6.389 g

A) 75% B) 56%
C) 65% D) 35%

Q10: When magnesium turnings are combusted (reacted with air), the by-product formed is ..... and is being removed using .....

A) Mg_2N_3/ NH_3 B) $Mg(OH)_2/H_2O$ C) Mg_3N_2/ NH_3 D) Mg_3N_2/ H_2O

Q11: Based on what you studied about Dumas method, calculate density of methanol (CH_3OH) (in g/dm^3) for a sample of methanol heated in a water bath at 42.0 °C and 0.37 atm. The mass of condensed vapor is 0.816 g.

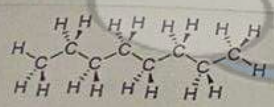
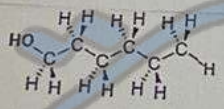
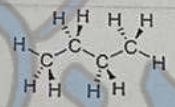
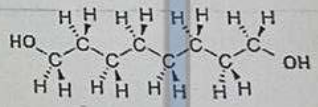
A) 0.93 B) 1.14 C) 0.46 D) 1.23

Q12: This equipment shown on the right is used during experiment of ..... to.....



A) Water of crystallization / cool the anhydrous salt away from humidity
B) Empirical formula / remove the undesired by-product.
C) Water of crystallization / keep the hydrated salt in a dry place.
D) Empirical formula / prevent reaction of magnesium with air.

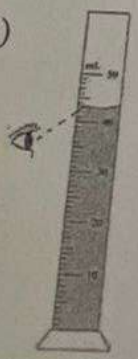
Q13: Which one of the following has the highest boiling point?

A)  B)  C)  D) 

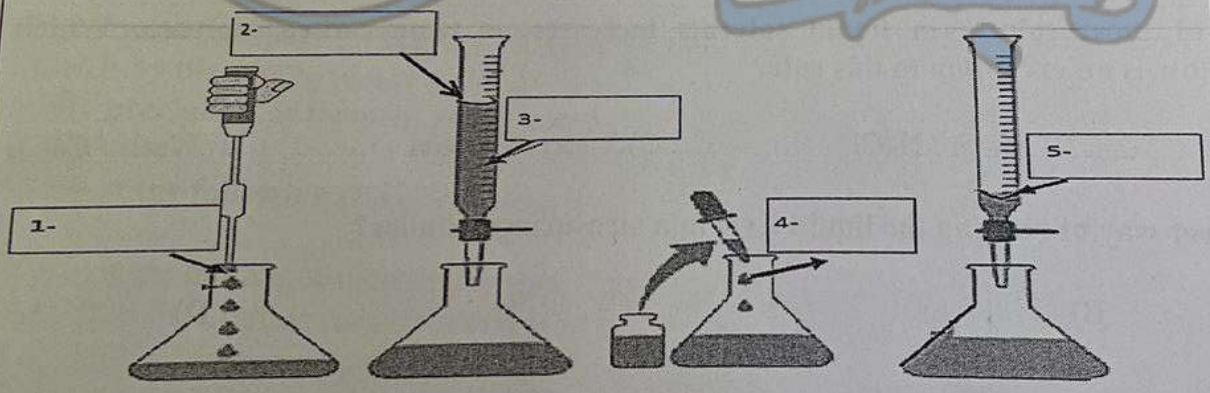
Q14: The solubility of most solutes in liquid solvents increases as temperature increase. Which solvent/solute listed below is an exception to this rule?

A) Water / Sugar B) Water / NaCl C) Water / Oxygen D) Water / $CaCl_2$

Q15: Which is the correct way of reading the liquid level in a measuring cylinder?

A)  B)  C)  D) 

PART2: Answer the following questions:

Q		Answer
1	Calculate ΔH° for the combustion reaction of butene: $2C_4H_{8(g)} + 12O_{2(g)} \rightarrow 8CO_{2(g)} + 8H_2O_{(l)}$, $\Delta H^\circ = ?$ Use the following data: 1) $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$ $\Delta H^\circ = -571$ kJ 2) $C_4H_{8(g)} + H_{2(g)} \rightarrow C_4H_{10(g)}$ $\Delta H^\circ = -126$ kJ 3) $2C_4H_{10(g)} + 13O_{2(g)} \rightarrow 8CO_{2(g)} + 10H_2O_{(l)}$ $\Delta H^\circ = -575$kJ	
2	The indicator used in Vinegar analysis is:	
3	When $CaCl_2$ solution is allowed to react with Na_2CO_3 solution, a white precipitate is formed. If the precipitate not collected completely, the percentage yield of the reaction will: (Increase – decrease – stay the same)	
4	Which liquid's molar mass could be determined by Dumas method? **Toluene (B.p 110°C) OR Acetone (B.p 56°C).**	
5	Give one sign or an observable change that indicates that water is completely removed when a hydrated salt is being heated.	
6	The method you use to find the empirical formula of magnesium oxide in your lab is called …….. method	
7	The reaction of neutralization of acid base is : (exothermic or endothermic)	
8	The volume of an object is 2.25 m^3 and its density = 2.55 kg/m^3. What is the mass of the object?	
9	The following picture shows an equipment called….. 	
10	What the following numbers indicate, including the name of the chemical (1-5) in Bleach titration…… 	

GOOD LUCK,,,

Al-Balqa Applied University

PART1: Choose the answer of the following question's and transfer it to the table in CAPITAL letters.

Q1: The specific heat of copper is 0.385 J/g.K . Find the amount of heat needed to raise up the temperature of a 1.42 kg block of copper from 25.0 °C to 159.3 °C?

A) 127.16 J B) 34.72×10^3J C) 358.75 J D) 73.42×10^3J

Q2: When 50.0 g of 0.200 M $NaCl_{(aq)}$ at 24.1 °C are added to 100.0 g of 0.100 M $AgNO_{3(aq)}$ in a cup calorimeter, the temperature raises up from 24.1 °C to 24.8 °C as $AgCl_{(s)}$ precipitate forms. Assuming the specific heat of the solution and products is 4.20 J/g °C. Calculate the approximate amount of heat in joules (J) produced by the reaction.

A) – 693 J B) – 441 J C) + 693 J D) + 441 J

Q3: Suppose a bleach solution is 8.75% by mass of sodium hypochlorite NaOCL. The mass of chlorine (Cl_2) in this solution is:

A) 4.17 g B) 5.23 g C) 6.85 g D) 8.33 g

Q4: The active ingredient in bleach is considered as a:

A) Weak reducer B) Weak oxidizer C) Strong oxidizer D) Strong reducer

Q5: What is the volume (in mL) of 0.586 M $Ba(OH)_2$ required to neutralize a 1.40 g of acetic acid dissolved in 5 mL water? (Balance the equation)

$$CH_3COOH_{(aq)} + Ba(OH)_{2(aq)} \longrightarrow Ba(CH_3COO)_{2(aq)} + H_2O_{(l)}$$

A) 19.9 mL B) 9.98 mL C) 12.5 mL D) 4.98 mL

Q6: Concerning the experiment of vinegar analysis, a student got a dark pink color at the endpoint of titration instead of the rose pink (light pink) color. How is the calculated Molarity of CH_3COOH in the vinegar sample affected, compared to the actual one?

A) Greater than actual B) Less than actual C) Same as actual D) Not affected

Q7: A student prepared a solution of $AgNO_3$ by dissolving (2.6 g $AgNO_3$) in (100 mL) water, and a solution of $CaCl_2$ by dissolving (0.629 g $CaCl_2$) in (85 mL) water. When 15 mL of each solution are mixed together a white precipitate is formed, and allowed to react according to the following equation:

$$2AgNO_3 + CaCl_2 \longrightarrow 2AgCl + Ca(NO_3)_2$$

Which of the following statements is correct?

A) $CaCl_2$ is the limiting reagent and it gives 0.002 mol of AgCl precipitate.
B) $AgNO_3$ is the limiting reagent and it gives 0.002 mol of AgCl precipitate.
C) $CaCl_2$ is the limiting reagent and it gives 0.001 mol of $Ca(NO_3)_2$ precipitate.
D) $AgNO_3$ is the limiting reagent and it gives 0.001 mol of $Ca(NO_3)_2$ precipitate.